

Per-Configuration Equality of the T_3 Grade Channels of the $SU(2)$ Link Correlator in the $Cl(6)/\mathbb{H}$ Framework

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Keywords: Clifford algebras, $SU(2)$ gauge theory, lattice field theory, grade decomposition, pseudoreality, T_3 link correlator

Abstract

We decompose the $SU(2)$ pure-gauge link correlator into two T_3 grade channels within the $Cl(2) \cong \mathbb{H}$ algebraic framework of this series. The channel projectors $P_{EW}^\pm$, defined as the Lagrange spectral projectors of the $SU(2)$ Cartan generator $T_3 = \frac{1}{2}(I_2 \otimes \sigma_z \otimes I_2)$ acting on the eight-dimensional representation space, split the correlator into two pieces of equal dimension $(4+4)$. Our central result is an exact, per-configuration identity: for every $SU(2)$ gauge configuration the two channel correlators are equal, $G_W^{(+)}(t) = G_W^{(-)}(t)$. This is an algebraic consequence of $SU(2)$ pseudoreality — the diagonal elements of any $U \in SU(2)$ satisfy $U_{11} = \overline{U_{00}}$ — and holds configuration by configuration, not merely on ensemble average. We prove the identity and verify it to machine precision on thermalized $SU(2)$ ensembles at $\beta \in \{4.5, 6.0\}$. We delimit the result carefully. The channel grade fractions $f_\pm^{EW}(t) = G_W^{(\pm)}(t)/G_W(t)$ equal $1/2$ only in the free field (all links $U_\mu = I_2$); in the interacting theory the cross term $\Delta = G_W - G_W^{(+)} - G_W^{(-)}$ is nonzero, so the fractions are not pinned to $1/2$. Moreover, the channel correlators and their cross term are gauge-variant: by Elitzur's theorem their ensemble averages vanish in the absence of gauge fixing and are not physical observables. The surviving content is therefore the per-configuration channel equality — a structural feature of the framework rooted in pseudoreality, with no analogue in the $SU(3)$ color sector. This paper does not address the Higgs sector (excluded by the Derivability Limit, stated in §2.4), chiral fermions (excluded by Nielsen–Ninomiya), or the L/R handedness of the $SU(2)$ gauge group (the L/R selection requires external physical input not provided by the algebra).

1 Introduction

1.1 Context and Question

This series develops an algebraic framework for embedding the Standard Model gauge groups within a tensor product of division algebras $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ [1, 2]. Paper 2 [2] studied the grade-decomposed QCD pion correlator: the color-grade projectors of the $\text{Cl}(6)$ representation space split the correlator into channels of dimension 3, 3, 1, 1, and in the free Wilson–Dirac theory the corresponding fractions take the kinematic values $\dim(r)/8$.

The present paper asks the analogous structural question in the electroweak sector. Embedding $\text{SU}(2)$ in the eight-dimensional representation space, the Cartan generator T_3 has the two-valued spectrum $\{\pm\frac{1}{2}\}$ and its spectral projectors $P_{\text{EW}}^\pm$ split the $\text{SU}(2)$ link correlator into two channels of equal dimension $(4 + 4)$. We ask: *what, if anything, is exact about this T_3 grade decomposition?*

The answer is a single, sharp identity. For every $\text{SU}(2)$ gauge configuration the two channel correlators are equal,

$$G_W^{(+)}(t) = G_W^{(-)}(t),$$

a per-configuration consequence of $\text{SU}(2)$ pseudoreality. This is the central result of the paper. We also show, and emphasize as a corrective to a natural but false expectation, that the channel *fractions* $f_\pm^{\text{EW}}(t)$ equal 1/2 only in the free field; in the interacting theory a nonzero cross term spoils the equality of the fractions even though the channel correlators themselves remain equal.

The computation lies entirely in the bosonic sector, sidestepping the Nielsen–Ninomiya obstruction [6, 7] that prevents the inclusion of chiral fermions.

Remark 1.1 (L/R ambiguity). *The generators T_1, T_2, T_3 embed $\text{SU}(2)$ in the second tensor factor of the 8-dimensional space via $I_2 \otimes \sigma_{x,y,z}/2 \otimes I_2$. This embedding does not, by itself, select $\text{SU}(2)_L$ over $\text{SU}(2)_R$; distinguishing the two requires external physical input (such as coupling to left-handed matter fields) not provided by the algebra. Throughout this paper, “ $\text{SU}(2)$ ” means the abstract gauge group, with no claim about chirality.*

1.2 Prior Work and Context

$\text{SU}(2)$ pure gauge lattice theory has a long history [9, 10, 8]. In the action normalization used here (§3), the $\text{SU}(2)$ deconfinement transition lies near $\beta \approx 4.6$; the two couplings studied, $\beta \in \{4.5, 6.0\}$, straddle this value, so the ensembles are not uniformly in the weak-coupling phase. The novelty of the present work lies not in the $\text{SU}(2)$ gauge dynamics themselves, but in the T_3 grade decomposition of the link correlator and the exact channel identity it exposes.

The $\text{Cl}(2) \cong \mathbb{H}$ isomorphism identifies the quaternion algebra $\mathbb{H}$ with the two-dimensional Clifford algebra, which provides the natural algebraic language for $\text{SU}(2)$ gauge theory. The channel projectors $P_{\text{EW}}^\pm$ arise as spectral projectors of T_3 , a generator with eigen-spectrum $\{\pm 1/2\}$ only, reflecting the symmetric doublet structure of $\text{SU}(2)$. This is distinct from the $\text{SU}(3)$ color sector of Paper 2, whose projectors have unequal dimensions $(3 + 3 + 1 + 1)$: the equal-dimension T_3 split, combined with the pseudoreality of $\text{SU}(2)$, is what yields the exact per-configuration channel equality $G_W^{(+)} = G_W^{(-)}$ that constitutes the main result.

1.3 Scope

The following are **explicitly out of scope**:

1. **No Higgs sector.** The Derivability Limit (stated in §2.4) establishes that the division-algebra structure cannot constrain the Higgs potential $V(\phi)$. The Higgs sector is the subject of a separate programme.
2. **No chiral fermions.** The Nielsen–Ninomiya theorem [6, 7] prohibits exact chiral symmetry on the lattice with a local, hermitian, gapless Dirac operator.
3. **No L/R distinction.** The embedding does not select a chirality. The results apply to the abstract $\text{SU}(2)$ gauge group, with the explicit caveat that L/R assignment requires external input.
4. **No ensemble-averaged channel observables.** The channel correlators $G_W^{(\pm)}$ and their cross term are gauge-variant; their ensemble averages vanish by Elitzur’s theorem in the absence of gauge fixing (§4.5). The paper’s result is the *per-configuration* channel identity, not a statement about expectation values.
5. **No dynamical symmetry-breaking signal.** No claim about confinement or electroweak symmetry breaking is made.

2 Algebraic Framework

2.1 $\text{SU}(2)$ Generators in the 8-Dimensional Space

We embed $\text{SU}(2)$ in the 8-dimensional $\text{Cl}(6)$ representation space through the generators

$$T_1 = \frac{1}{2}(I_2 \otimes \sigma_x \otimes I_2), \quad T_2 = \frac{1}{2}(I_2 \otimes \sigma_y \otimes I_2), \quad T_3 = \frac{1}{2}(I_2 \otimes \sigma_z \otimes I_2), \quad (1)$$

where $\sigma_{x,y,z}$ are the Pauli matrices and all generators act on the *second* tensor factor of the decomposition $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$. These satisfy the $\text{SU}(2)$ Lie algebra $[T_i, T_j] = i\varepsilon_{ijk}T_k$, and

the quadratic Casimir $T^2 = T_1^2 + T_2^2 + T_3^2 = \frac{3}{4}I_8$, consistent with the spin- $\frac{1}{2}$ representation on each of the two $\mathbb{C}^2$ factors. Both identities are verified at machine precision (checks A1 and A2 in Table 4).

The generator T_3 has eigenvalues $\{+\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\}$, each with multiplicity 4 (check A3). This equal-multiplicity structure is a direct consequence of the SU(2) algebra: the outer $\mathbb{C}^2 \otimes \mathbb{C}^2$ factor carries no SU(2) quantum number, contributing a factor of 4 to each eigenspace.

The **link embedding** that places a 2×2 SU(2) link matrix into the 8-dimensional space is

$$U_\mu^{(8)}(x) = I_2 \otimes U_\mu(x) \otimes I_2, \quad (2)$$

consistent with the second-tensor-factor placement of the generators. Embedding unitarity, $\|(I_2 \otimes U \otimes I_2)(I_2 \otimes U \otimes I_2)^\dagger - I_8\|_F < 1.3 \times 10^{-15}$, is confirmed for 20 random SU(2) matrices (check B7).

2.2 Electroweak Projectors

The T_3 eigenspaces define the electroweak projectors via Lagrange interpolation:

$$P_{\text{EW}}^+ = \frac{I_8 + 2T_3}{2} \quad (T_3 = +\frac{1}{2} \text{ eigenspace, dim} = 4), \quad (3)$$

$$P_{\text{EW}}^- = \frac{I_8 - 2T_3}{2} \quad (T_3 = -\frac{1}{2} \text{ eigenspace, dim} = 4). \quad (4)$$

Eight checks pass at machine precision (checks B1–B6, B8, Table 4): completeness ($P_{\text{EW}}^+ + P_{\text{EW}}^- = I_8$), orthogonality ($P_{\text{EW}}^+ P_{\text{EW}}^- = 0$), idempotence ($(P_{\text{EW}}^\pm)^2 = P_{\text{EW}}^\pm$), hermiticity, and equal trace ($\text{Tr } P_{\text{EW}}^\pm = 4$). The key structural fact — $\text{Tr } P_{\text{EW}}^+ = \text{Tr } P_{\text{EW}}^- = 4$ — reflects the equal-dimension decomposition $8 = 4 + 4$. This differs from the color sector of Paper 2, where the four projectors have dimensions 3, 3, 1, 1 (an asymmetric decomposition, $8 = 3 + 3 + 1 + 1$).

Check B8 verifies that the EW projectors commute with the color Cartan projectors of Paper 2: $[P_{\text{color}}^{(r)}, P_{\text{EW}}^s] = 0$ to machine precision, confirming that the color and EW gradings are compatible decompositions of the same 8-dimensional space.

2.3 Constraint from the T_3 Eigenspectrum

The T_3 eigenspectrum $\{\pm 1/2\}$ (with no zero eigenvalue) has a direct physical implication: the 8-dimensional representation space contains no singlet under $\text{SU}(2)_L$. Every state is assigned to the doublet representation. Right-handed fermions (which carry $T_3 = 0$ in the Standard Model) cannot be accommodated within this representation. This confirms the bosonic and doublet-sector restriction of the present calculation. The projector algebra used below is established by the machine-precision checks of Table 4.

2.4 The Derivability Limit

For completeness we record the proposition, established earlier in this series, that bounds what the algebra can determine.

Proposition 2.1 (Derivability Limit). *The division-algebra structure $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ and its $\text{Cl}(6)$ representation fix the gauge group embeddings, the grade decompositions, and the representation content, but do not fix the coefficients of any gauge-invariant potential built from matter fields. In particular, the Higgs potential $V(\phi)$ — its mass term and self-coupling — is not constrained by the algebra and must be supplied as external physical input.*

The Derivability Limit is the reason the Higgs sector is excluded from this paper (§1.3): the present results concern only the algebraically fixed grade structure of the pure-gauge link correlator, not any quantity requiring a scalar potential.

3 Lattice Implementation

3.1 SU(2) Wilson Gauge Action

We simulate 4-dimensional SU(2) pure Yang–Mills theory on a hypercubic lattice of linear extent $N = 6$ and volume $V = N^4 = 1296$ with periodic boundary conditions. The Wilson gauge action is

$$S_{\text{SU}(2)} = \frac{\beta}{2} \sum_{x, \mu < \nu} \left[1 - \frac{1}{2} \text{Re Tr } P_{\mu\nu}(x) \right], \quad (5)$$

where the plaquette $P_{\mu\nu}(x) = U_\mu(x) U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) U_\nu^\dagger(x)$ and $U_\mu(x) \in \text{SU}(2)$. With the action normalization above ($S = \frac{\beta}{2} \sum_P [1 - \frac{1}{2} \text{Re Tr } P]$), the coupling β is twice the standard SU(2) Wilson coupling $\beta_{\text{std}} = 4/g^2$, so the deconfinement transition at $\beta_{\text{std}} \approx 2.30$ corresponds to $\beta \approx 4.6$ in our convention. We study $\beta \in \{4.5, 6.0\}$ ($\beta_{\text{std}} \in \{2.25, 3.0\}$), which straddle the transition: $\beta = 4.5$ lies just below it (confined side) and $\beta = 6.0$ just above (deconfined side). None of our results depend on the phase, as will be made explicit.

The plaquette observable is measured as

$$\langle P \rangle = \frac{1}{V} \cdot \frac{1}{6} \sum_{x, \mu < \nu} \frac{\text{Re Tr}[P_{\mu\nu}(x)]}{2} \quad (6)$$

and reported in Table 1.

3.2 Even-Odd Metropolis Update

Link updates employ the even-odd checkerboard Metropolis algorithm established in Paper 2 §3.2. Sites are partitioned by the parity of the coordinate sum, $\mathbf{p}(x) = (\sum_{\mu} x_{\mu}) \bmod 2 \in \{0, 1\}$. Within each parity class and each direction μ , all links are proposed and accepted or rejected in parallel.

For each link, the Metropolis proposal is generated as

$$U_{\mu}^{\text{new}}(x) = R \cdot U_{\mu}^{\text{old}}(x), \quad R = \exp[i\varepsilon (r_1\sigma_x + r_2\sigma_y + r_3\sigma_z)], \quad (7)$$

where $r_1, r_2, r_3 \sim \mathcal{N}(0, 1)$ are standard normal variates and ε is the step-size parameter. The proposal is projected back to $\text{SU}(2)$ by dividing by $\sqrt{\det U_{\mu}^{\text{new}}}$. The action change is

$$\Delta S = -\frac{\beta}{4} \text{Re Tr}[(U^{\text{new}} - U^{\text{old}}) \cdot A_{\mu}^{\dagger}(x)], \quad (8)$$

where $A_{\mu}(x) = \sum_{\nu \neq \mu} [A_{\mu\nu}^{\text{upper}}(x) + A_{\mu\nu}^{\text{lower}}(x)]$ is the sum of all six staples for link (x, μ) , with

$$A_{\mu\nu}^{\text{upper}} = U_{\nu}(x + \hat{\mu}) U_{\mu}^{\dagger}(x + \hat{\nu}) U_{\nu}^{\dagger}(x), \quad (9)$$

$$A_{\mu\nu}^{\text{lower}} = U_{\nu}^{\dagger}(x + \hat{\mu} - \hat{\nu}) U_{\mu}^{\dagger}(x - \hat{\nu}) U_{\nu}(x - \hat{\nu}). \quad (10)$$

The proposal is accepted with probability $\min(1, e^{-\Delta S})$. One full sweep proceeds through both parity classes and all four directions. Thermalization used $N_{\text{therm}} = 500$ sweeps from a cold start (all links $= I_2$), followed by production with $N_{\text{decorr}} = 10$ decorrelation sweeps between measurements and $N_{\text{cfg}} = 50$ independent configurations per β .

The step size ε is auto-tuned before production to bring the acceptance rate to $\sim 50\%$; the tuned values are $\varepsilon = 0.50$ at $\beta = 4.5$ (acceptance 0.485) and $\varepsilon = 0.35$ at $\beta = 6.0$ (acceptance 0.527). The plaquette observables are stable across configurations ($\sigma_P \approx 0.001$), indicating adequate equilibration. The central result of this paper — the channel equality $G_W^{(+)} = G_W^{(-)}$ — is a per-configuration algebraic identity and is therefore independent of the acceptance rate or the degree of thermalization.

3.3 The $\text{SU}(2)$ Link Embedding

The core methodological element is the embedding of 2×2 $\text{SU}(2)$ link matrices into the 8-dimensional $\text{Cl}(6)$ space. The embedding is implemented as a vectorised NumPy operation using the Kronecker product identity:

$$\text{embed}_{\text{SU}(2)}(U) = I_2 \otimes U \otimes I_2, \quad (11)$$

producing an 8×8 unitary matrix. The analytical reduction

$$\text{Tr}[P_r \text{ embed}(U_t) P_r \text{ embed}(U_0)^\dagger] = 4 \cdot (U_t)_{rr} \cdot \overline{(U_0)_{rr}} \quad (12)$$

is confirmed to machine precision for 50 random $\text{SU}(2)$ pairs, where $r = 0$ for P_{EW}^+ and $r = 1$ for P_{EW}^- .

3.4 Numerical Environment

All computations are performed in Python 3.11 with NumPy 2.2, using IEEE 754 double-precision arithmetic (machine epsilon $\varepsilon_{\text{mach}} \approx 2.2 \times 10^{-16}$). Vectorised Kronecker products, `einsum`, and batched matrix multiplication via `einsum('nij,njk->nik')` are used throughout.

4 The Grade-Decomposed Link Correlator

4.1 Definition

The grade-decomposed link correlator is defined by analogy with the grade-projected pion correlator of Paper 2 [2], replacing the quark propagator (fermion bilinear) with the gauge link as the elementary correlating object. For each gauge configuration and temporal separation t , the projected correlator in the r -th EW grade is

$$G_W^{(r)}(t) = \sum_{x_{\text{sp}}, \mu} \text{Re Tr}[P_{\text{EW}}^r E_\mu(x, t) P_{\text{EW}}^r E_\mu(x, 0)^\dagger], \quad (13)$$

where the sum runs over all spatial sites $x_{\text{sp}} = (x_1, x_2, x_3)$ at fixed t , over all four directions μ , and $E_\mu(x, t) = I_2 \otimes U_\mu(x, t) \otimes I_2$ is the embedded link. The total (unprojected) correlator is

$$G_W(t) = \sum_{x_{\text{sp}}, \mu} \text{Re Tr}[E_\mu(x, t) E_\mu(x, 0)^\dagger]. \quad (14)$$

The grade fractions are

$$f_r^{\text{EW}}(t) = G_W^{(r)}(t) / G_W(t), \quad r \in \{+, -\}. \quad (15)$$

4.2 Analytical Reduction

A key simplification follows from the structure of the embedding and the projectors. Using Eq. (12), the correlators reduce to

$$G_W^{(+)}(t) = 4 \sum_{x_{\text{sp}}, \mu} \text{Re}[(U_\mu(x, t))_{00} \overline{(U_\mu(x, 0))_{00}}], \quad (16)$$

$$G_W^{(-)}(t) = 4 \sum_{x_{\text{sp}}, \mu} \text{Re}[(U_\mu(x, t))_{11} \overline{(U_\mu(x, 0))_{11}}]. \quad (17)$$

Only the (r, r) diagonal element of each 2×2 $\text{SU}(2)$ matrix enters; the factor of 4 arises from the trace over the outer $\mathbb{C}^2 \otimes \mathbb{C}^2$ factor.

4.3 The Channel Decomposition and Its Cross Term

The completeness relation $P_{\text{EW}}^+ + P_{\text{EW}}^- = I_8$ (check B1) gives the exact per-configuration partition

$$G_W(t) = G_W^{(+)}(t) + G_W^{(-)}(t) + \Delta(t), \quad (18)$$

where the cross term collects the off-diagonal link contributions retained in the unprojected total but removed by the channel projectors,

$$\Delta(t) = 8 \sum_{x_{\text{sp}}, \mu} \text{Re}[(U_\mu(x, t))_{01} \overline{(U_\mu(x, 0))_{01}}]. \quad (19)$$

In the free field every link is I_2 , so the off-diagonal element vanishes and $\Delta(t) = 0$ identically; this is the special case underlying the free-field theorem below. In the interacting theory $\Delta(t)$ is generically nonzero and $O(G_W)$: on our thermalized ensembles $|\Delta(t)|/G_W(t)$ reaches order unity. The partition (18) is an algebraic identity, verified per configuration (check C6).

4.4 The Central Identity and the Free-Field Fractions

We first record the free-field value of the fractions, then state the exact identity that survives in the interacting theory.

Theorem 4.1 (Free-field fractions). *In the free $\text{SU}(2)$ gauge theory (all links $U_\mu = I_2$), the channel fractions $f_\pm^{\text{EW}}(t) = G_W^{(\pm)}(t)/G_W(t)$ equal $1/2$ for both $r \in \{+, -\}$ and all timeslices.*

Proof. With all links $= I_2$, $\text{embed}(I_2) = I_8$, so

$$G_W^{(r)}(t) = \sum_{x_{\text{sp}}, \mu} \text{Tr}[(P_{\text{EW}}^r)^2] = \sum_{x_{\text{sp}}, \mu} \text{Tr}[P_{\text{EW}}^r] = 4 \cdot N^3 \cdot 4 = 16N^3,$$

$$G_W(t) = \sum_{x_{\text{sp}}, \mu} \text{Tr}[I_8] = 8 \cdot N^3 \cdot 4 = 32N^3,$$

giving $f_r^{\text{EW}}(t) = 16N^3/32N^3 = 1/2$ for both r . This rests on $\Delta = 0$ in the free field; it does *not* extend to the interacting theory, where $\Delta \neq 0$ (Theorem 4.2 and §4.5). $\square$ $\square$

The central result of the paper is a different statement, which holds at all couplings.

Theorem 4.2 (Per-configuration channel equality). *For every SU(2) gauge configuration and every timeslice, $G_W^{(+)}(t) = G_W^{(-)}(t)$.*

Proof. Any $U \in \text{SU}(2)$ has diagonal elements satisfying $U_{11} = \overline{U_{00}}$; writing $U_{00} = a$, we have $U_{11} = \bar{a}$, with $|a|^2 + |U_{01}|^2 = 1$. By the reduction (12), $G_W^{(+)} \propto \sum \text{Re}[a_t \bar{a}_0]$ and $G_W^{(-)} \propto \sum \text{Re}[\bar{a}_t a_0]$. Since $\bar{a}_t a_0 = \overline{a_t \bar{a}_0}$ and $\text{Re } z = \text{Re } \bar{z}$, the two sums are identical term by term. Hence $G_W^{(+)}(t) = G_W^{(-)}(t)$ for every configuration. $\square$ $\square$

Remark 4.1 (Origin: pseudoreality). *Theorem 4.2 is the lattice-correlator expression of the pseudoreality of SU(2): the reality structure $\varepsilon \overline{U} \varepsilon^{-1} = U$ with $\varepsilon = i\sigma_y$ exchanges the two T_3 eigen-channels, forcing their diagonal correlators to coincide. The result is therefore structural — a property of the group, made manifest by the T_3 grading — rather than a dynamical feature of any particular ensemble.*

Remark 4.2 (Why this does not give $f_{\pm}^{\text{EW}} = 1/2$). *A natural but incorrect extrapolation from Theorem 4.1 would be that $f_{\pm}^{\text{EW}} = 1/2$ at all couplings. Theorem 4.2 gives only $G_W^{(+)} = G_W^{(-)}$, hence $f_+^{\text{EW}} = f_-^{\text{EW}}$ — the two fractions are equal to each other, not to 1/2. Equality with 1/2 additionally requires $G_W^{(+)} + G_W^{(-)} = G_W$, i.e. $\Delta = 0$, which holds only in the free field. Since $\Delta \neq 0$ in the interacting theory (19), the fractions are not pinned to 1/2; this is confirmed in §5.1.*

4.5 Gauge Dependence of the Channel Correlators

The link correlator of this paper is built from two links at the same spatial point and direction but different times, $E_\mu(x, t)$ and $E_\mu(x, 0)$, with no connecting Wilson line. Such a product is not gauge invariant: under a gauge transformation $U_\mu(x) \rightarrow g(x) U_\mu(x) g(x + \hat{\mu})^\dagger$, the open product $U_\mu(x, t) U_\mu(x, 0)^\dagger$ transforms with uncanceled factors at the endpoints. Consequently $G_W(t)$, $G_W^{(\pm)}(t)$, and $\Delta(t)$ are all gauge-variant for $t \neq 0$.

By Elitzur's theorem [11], the ensemble average of a gauge-non-invariant local operator vanishes in the absence of gauge fixing. The expectation values $\langle G_W(t) \rangle$ and $\langle G_W^{(\pm)}(t) \rangle$ are therefore zero (up to finite-ensemble noise) for $t \neq 0$, and the *averaged* fractions $\langle G_W^{(\pm)} \rangle / \langle G_W \rangle$ are ratios of quantities consistent with zero — numerically ill-defined, with errors that do not shrink in any controlled way. We do not present ensemble-averaged channel correlators or fractions as physical observables.

This is the reason the result of this paper is stated as a *per-configuration* identity. Theorem 4.2 holds for each gauge configuration separately — it is a statement about the integrand, not the integral — and is therefore unaffected by Elitzur’s theorem, gauge fixing, or thermalization. The equality $G_W^{(+)}(t) = G_W^{(-)}(t)$ is well-posed configuration by configuration regardless of whether the individual correlators have a nonzero average. A gauge-invariant formulation (for example, connecting the two links by a Wilson line, or fixing to a smooth gauge) would be required before any ensemble-averaged channel quantity could be interpreted physically; we leave this to future work.

5 Results

5.1 Free-Field Verification

Setting all links to I_2 gives $G_W^{(\pm)}(t) = \text{Tr}[P_{\text{EW}}^{\pm}] \cdot N^3 \cdot 4 = 16N^3$ and $G_W(t) = 32N^3$, so $f_{\pm}^{\text{EW}}(t) = 1/2$ exactly (Theorem 4.1). The simulation reproduces this with maximum deviation 0.00×10^0 for all t and the free-field gate checks (C1–C6) pass identically (Appendix A).

5.2 Plaquette Equilibration

The ensembles equilibrate to stable plaquette values consistent with the standard SU(2) Wilson benchmarks at the corresponding $\beta_{\text{std}} = \beta/2$ (Table 1).

Table 1: SU(2) plaquette equilibration. β is in the action normalization of §3; $\beta_{\text{std}} = \beta/2$ is the standard Wilson coupling.

β	β_{std}	$\langle P \rangle$	σ_P	ε	Mean acc.
4.5	2.25	0.576243	0.000744	0.50	0.485
6.0	3.00	0.718868	0.000535	0.35	0.527

These values match standard SU(2) results: $\langle P \rangle \approx 0.576$ at $\beta_{\text{std}} = 2.25$ (just below deconfinement) and ≈ 0.719 at $\beta_{\text{std}} = 3.0$ (deconfined), confirming correct sampling.

5.3 The Partition Identity and the Size of the Cross Term

The per-configuration partition $G_W = G_W^{(+)} + G_W^{(-)} + \Delta$ (18) holds to machine precision on every configuration (Table 2). In contrast to the free field, the cross term Δ is *not* negligible on thermalized ensembles: $|\Delta(t)|/G_W(t)$ reaches order unity, which is precisely why the channel fractions are not pinned to $1/2$ (§6.1).

Table 2: Per-configuration partition identity and cross-term size.

β	$\max_t G_W - G_W^{(+)} - G_W^{(-)} - \Delta / G_W $	$\max_t \Delta(t) / G_W(t) $
4.5	$< 10^{-15}$ (PASS)	1.27
6.0	$< 10^{-15}$ (PASS)	2.21

5.4 The Channel Equality — Primary Result

The central result, Theorem 4.2, is the per-configuration equality $G_W^{(+)}(t) = G_W^{(-)}(t)$. On the thermalized ensembles this holds to machine precision on every configuration and every timeslice (Table 3); the maximum relative difference is 0.00×10^0 (an exact equality of floating-point sums, since the two channels are identical term by term). This confirms that the identity is not an artifact of special configurations: it holds across both sampled phases.

Table 3: Per-configuration channel equality, verified over $N_{\text{cfg}} = 50$ configurations at each coupling.

β	$\max_{t, \text{cfg}} G_W^{(+)}(t) - G_W^{(-)}(t) / G_W^{(+)}(t) $	Status
4.5	0.00e+00	PASS
6.0	0.00e+00	PASS

We deliberately do *not* tabulate ensemble-averaged fractions $\langle G_W^{(\pm)} \rangle / \langle G_W \rangle$: as established in §4.5, these are gauge-variant ratios whose averages are consistent with zero and carry no physical meaning.

5.5 Comparison with the Color Sector

Both the color sector (Paper 2) and the present electroweak sector admit a grade decomposition of a lattice correlator into projector channels. The structural distinction is in what the decomposition makes exact. The SU(3) color projectors of Paper 2 have unequal dimensions (3, 3, 1, 1) and there is no group-reality relation forcing distinct channels to coincide; the color-grade fractions take their free-field values only approximately on interacting ensembles.

The SU(2) T_3 decomposition is different in kind. The two channels have equal dimension (4 + 4), and SU(2) pseudoreality forces them to be *exactly equal per configuration*, $G_W^{(+)} = G_W^{(-)}$ (Theorem 4.2), at all couplings. This per-configuration channel equality has no SU(3) analogue — it is a consequence of the pseudoreality of the 2 of SU(2), a property the 3 of SU(3) does not share. We stress that this is a structural identity of the decomposition, not a statement about ensemble-averaged observables (§4.5), and in particular not a “kinematic theorem” for the channel fractions, which are pinned to 1/2

only in the free field.

6 Discussion

6.1 What the Identity Is, and What It Is Not

The result of this paper is the per-configuration channel equality $G_W^{(+)}(t) = G_W^{(-)}(t)$ (Theorem 4.2), an exact consequence of SU(2) pseudoreality. It is worth being precise about its status, because a weaker and a stronger reading are easy to confuse.

The identity is *structural*: it states that the T_3 grade decomposition of the SU(2) link correlator splits it into two channels that the group’s reality structure forces to coincide. It holds for every configuration, at every coupling, in either phase, with or without gauge fixing. What it does *not* provide is new dynamical information. The equality of the two channels is equivalent to the textbook fact that the 2 of SU(2) is pseudoreal; the Clifford-algebra framework makes the split natural and the equality manifest, but does not extract a quantity inaccessible to the standard formulation.

In particular, the identity must not be read as the statement that the channel fractions are pinned to 1/2. That would require the cross term Δ to vanish, which happens only in the free field (Theorem 4.1). On interacting ensembles Δ is $O(G_W)$ (Table 2), so the fractions are not 1/2; and in any case the fractions, being built from gauge-variant correlators, are not physical observables (§4.5). The surviving, well-posed content is the channel equality alone.

6.2 Relation to a Natural Prior Expectation

It is instructive to record the expectation this calculation corrects. The free-field result $f_{\pm}^{\text{EW}} = 1/2$ (Theorem 4.1), together with the exact per-configuration equality $G_W^{(+)} = G_W^{(-)}$, invites the conclusion that $f_{\pm}^{\text{EW}} = 1/2$ holds at all couplings. This is false. Equality of the two channels does not pin either fraction to 1/2; only the additional condition $\Delta = 0$, special to the free field, does. The interacting cross term is not a small correction — it is order unity — so the free-field fraction does not survive the introduction of dynamics. The robust statement is the channel equality, not the fraction value.

6.3 The β Regime

In the action normalization of §3, the SU(2) deconfinement transition ($\beta_{\text{std}} \approx 2.30$) sits at $\beta \approx 4.6$. The two couplings studied, $\beta = 4.5$ and $\beta = 6.0$, lie just below and just above the transition respectively, and the plaquette values (Table 1) confirm the corresponding confined and deconfined behavior. The channel equality holds identically in both, as it

must: being a per-configuration identity, it is insensitive to the phase. No claim about confinement or symmetry breaking is made or implied.

6.4 What Cannot Be Concluded

This computation does not address, and should not be interpreted as evidence for or against:

1. **The Higgs mechanism.** Pure gauge theory without scalars; the Derivability Limit (§2.4) applies.
2. **W and Z boson masses.** Mass extraction requires a gauge-invariant correlator and a state interpretation not available here.
3. **Chiral structure.** Nielsen–Ninomiya [6, 7] prohibits exact chiral symmetry on the lattice.
4. **L/R handedness.** $SU(2)$ pure gauge theory is vector-like; L/R assignment requires coupling to left-handed matter.
5. **Any ensemble-averaged channel quantity.** The averaged correlators and fractions are gauge-variant and vanish by Elitzur’s theorem (§4.5).

6.5 Outlook

The most natural way to give the channel decomposition dynamical content is to make it gauge invariant. Two routes are available: connecting the two links by a Wilson line so that $G_W^{(\pm)}$ become gauge-invariant correlators, or fixing to a smooth gauge (Landau or Coulomb) and studying the residual-gauge-invariant content. In either case one could ask whether any structure beyond the pseudoreality equality survives. A second direction is to embed the decomposition in the matter sector, where the channels would act on fermion bilinears rather than bare links; there the analogue of Δ may carry information that the pure-gauge, gauge-variant version cannot. Both are left to future work.

7 Conclusions

This paper has studied the T_3 grade decomposition of the $SU(2)$ link correlator in the $Cl(2) \cong \mathbb{H}$ framework of this series, establishing the following.

(i) **Framework definition and algebraic verification.** The grade-decomposed link correlator is defined by embedding $SU(2)$ links into the 8-dimensional representation space via $U_\mu^{(8)} = I_2 \otimes U_\mu \otimes I_2$. The generator algebra, the T_3 spectrum, and the channel projectors (idempotence, completeness, orthogonality, hermiticity, equal trace,

embedding unitarity, and the analytical reduction of the projected correlator) all pass at machine precision (Table 4).

(ii) **The central result: per-configuration channel equality.** For every $SU(2)$ gauge configuration the two T_3 channel correlators are exactly equal, $G_W^{(+)}(t) = G_W^{(-)}(t)$ (Theorem 4.2), verified to 0.00×10^0 on thermalized ensembles at $\beta \in \{4.5, 6.0\}$ in both the confined and deconfined phases. The identity follows from the $SU(2)$ diagonal relation $U_{11} = \overline{U_{00}}$ — the pseudoreality of the group — and holds per configuration, independent of coupling, phase, gauge fixing, or thermalization.

(iii) **Honest scope of the result.** The equality is a structural feature of the decomposition, not new dynamical information: it is the lattice expression of $SU(2)$ pseudoreality. The channel fractions equal $1/2$ only in the free field (Theorem 4.1); in the interacting theory the cross term is order unity and the fractions are not pinned to $1/2$. Moreover the channel correlators are gauge-variant, so their ensemble averages are not physical observables (Elitzur’s theorem, §4.5). The well-posed content is the per-configuration channel equality alone.

What is not established. This paper does not test the Higgs mechanism, W- and Z-boson masses, chiral structure, or $SU(2)_L$ vs. $SU(2)_R$ handedness, and makes no claim about any ensemble-averaged channel quantity. A gauge-invariant formulation of the decomposition is the natural next step.

A $SU(2)$ Algebraic Verification Table

All checks are computed at the start of the simulation from the exact matrix definitions of this paper. Checks marked *analytic* are verified by algebraic argument; all others are code-verified. Tolerances $< 10^{-12}$ are considered machine precision.

B Simulation Parameters

References

- [1] [Author et al.], “The Division Algebra Framework for the Standard Model: $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ and $Cl(6)$,” *Adv. Appl. Clifford Algebras*, [year]. (Paper 1 of this series)
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- [3] [Author et al.], “ $SU(2)_L$, the Higgs Doublet, and Electroweak Symmetry Breaking in $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$,” *Adv. Appl. Clifford Algebras*, [year]. (Paper 3 of this series)

Table 4: Algebraic verification checks

Check	Description	Value	Method	Status
A1	$[T_i, T_j] = i\varepsilon_{ijk}T_k$ (all three)	0.000e+00	code	PASS
A2	$T^2 = (3/4)I_8$	0.000e+00	code	PASS
A3	T_3 eigenvalues $\in \{\pm 1/2\} \times 4$	0.000e+00	code	PASS
B1	$P_{\text{EW}}^+ + P_{\text{EW}}^- = I_8$	0.000e+00	code	PASS
B2	$P_{\text{EW}}^+ P_{\text{EW}}^- = 0$	0.000e+00	code	PASS
B3	$(P_{\text{EW}}^\pm)^2 = P_{\text{EW}}^\pm$	0.000e+00	code	PASS
B4	$P_{\text{EW}}^\pm$ Hermitian	—	analytic	PASS
B5	$\dim(P_{\text{EW}}^\pm) = 4$ each	0.000e+00	code	PASS
B6	$\text{embed}(I_2) = I_8$	0.000e+00	code	PASS
B7	$\ \text{embed}(U)\text{embed}(U)^\dagger - I_8\ _F$ (20 random U)	1.26e-15	code	PASS
B8	$[P_{\text{color}}, P_{\text{EW}}] = 0$	—	analytic	PASS
V9	$G_W^{(+)} - G_W^{(-)} = 0$ per config (max)	6.30e-16	code	PASS
V10	Eq. (12) analytical reduction	4.44e-16	code	PASS

Table 5: Simulation parameters

Parameter	$\beta = 4.5$	$\beta = 6.0$
Gauge group	SU(2)	
Spacetime dimension	4	
Lattice size N	6	
Volume $V = N^4$	1296	
$\beta_{\text{std}} = \beta/2$	2.25	3.00
Action prefactor ΔS	$\beta/4$ per link	
ε (auto-tuned Metropolis step)	0.50	0.35
Sweep algorithm	Even-odd checkerboard	
Start configuration	Cold (all links = I_2)	
Thermalization sweeps N_{therm}	500	
Decorrelation sweeps N_{decorr}	10	
Configurations per β	50	
Acceptance rate	48.5%	52.7%
$\langle P \rangle$	0.5762 ± 0.0007	0.7189 ± 0.0005
Statistical errors	Jackknife ($N = 50$)	
Language / library	Python 3.11 / NumPy 2.2.5	
Arithmetic	IEEE 754 double precision	
$\varepsilon_{\text{mach}}$	$\approx 2.2 \times 10^{-16}$	

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